



Course Projects

Proposal



Final Project Proposal Guidelines

Due: Friday, March 13

The proposal is a concise, one-page document that outlines your plan for the final project. The goal is to demonstrate that you have a clear research question, an appropriate dataset, and a well-justified analytical strategy using methods from this course.

This is not a literature review. It is a structured plan.

Proposal Must Include:

1. Project Title and Team Members

- List all team members (if working in a group -- work must scale!).
- Clearly identify one primary contact person.

2. Research Question

- State a focused, well-defined research question.
- Explain why the question is meaningful and appropriate for a data mining/statistical learning framework.

- If applicable, clearly state hypotheses.

Your question should require unsupervised learning. Simple summary statistics or tool development are not sufficient.

3. Dataset Description

- Describe the dataset you plan to use.
- Source (public repository, curated dataset, simulated data with justification, etc.).
- Number of observations and variables (if known).
- Any anticipated data challenges (missingness, imbalance, high dimensionality, etc.).

You must either already have access to the data or clearly state how you will obtain it.

4. Analytical Plan

Provide a structured outline of the methods you plan to use and why they are appropriate.

- Data preprocessing steps (cleaning, transformation, encoding, scaling, feature engineering, etc.)
- Unsupervised learning methods and why they are relevant.
- Model evaluation/performance measure strategy

The emphasis here is justification. Explain your reasoning—not just a list of techniques.

5. Expected Contributions / Insights

- What you expect to learn.
- What meaningful insight would look like.
- Potential limitations or challenges.

Important Notes

- Limited to one page (single-spaced, 11–12 pt font).
- Be concise and precise.
- Vague proposals will be returned for revision.
- You must meet with the instructor after Spring Break to finalize the project plan. You will be given an appointment.

Project Proposal



Assignment



Due March 13 at 11:59 PM



